

RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

Academic Year 2020-2021 (Odd Semester)

Innovative Practices(s) Description

Name of the Faculty	: R. Venkatesh, AP/Mechanical
Subject Code and Title	: ME8593– Design of Machine Elements
Year and Branch	: III –Mechanical Engineering
Topics Covered	: Universal Joint
Name of the Laboratory	: Mechanics and Machines Laboratory
Website Link	: http://mm-nitk.vlabs.ac.in/exp21/index.html
Date & Time	: 03.09.2020 & 10.40AM to 11.40AM

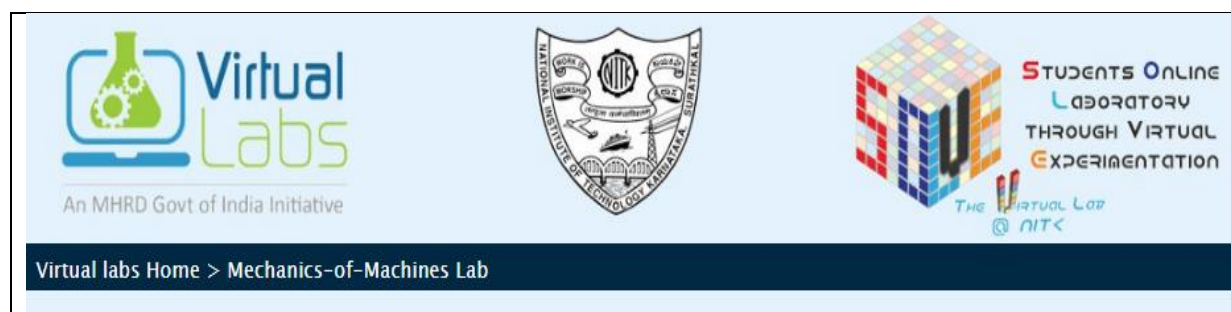
Demonstration of Universal Joint

OBJECTIVE:

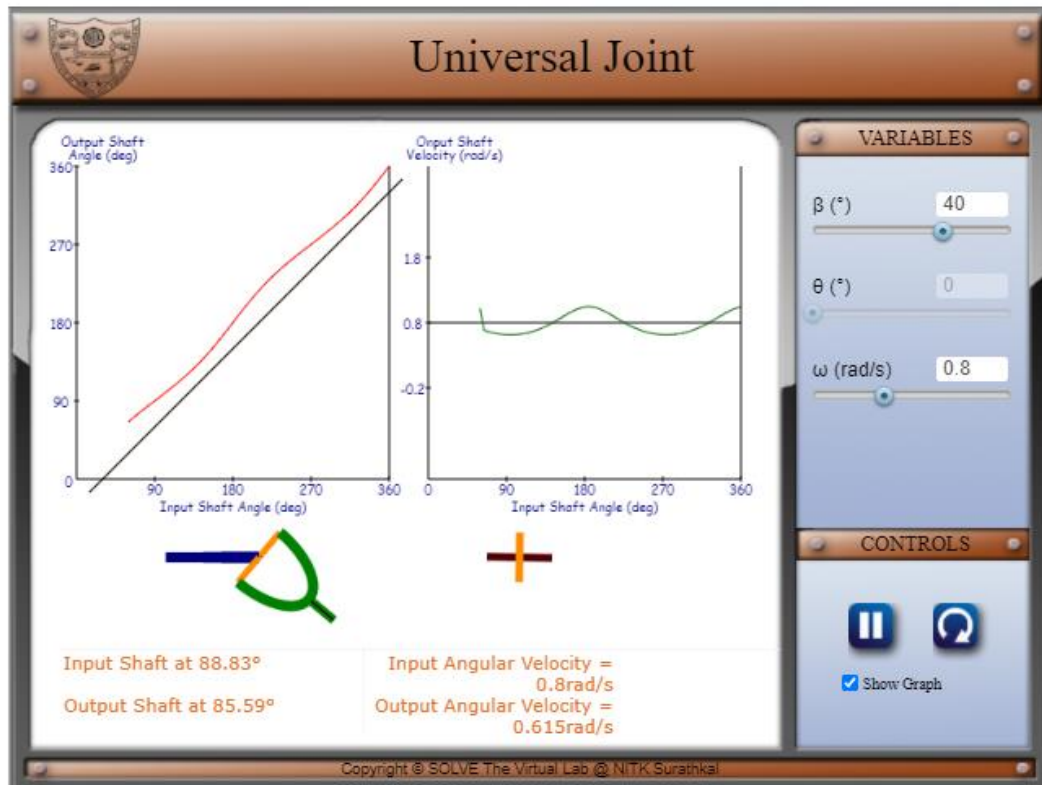
1. To understand the relative motion between the output and the input shaft of the universal joint.
2. To understand the variation of angle output shaft and input shaft.

Universal joint is a positive mechanical joint used for connecting shafts, whose axes are inclined at an angle to each other. It is also known as universal coupling, U-joint, Cardan Joint and Hooke's Joint. It compensates angular misalignment between the shafts in any direction.

The students can able to calculate the bend angle and angle of rotation of all the axle systems.



Simulation



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OUTCOME: The Virtual Laboratory Experiments helps in attaining the following CO-PO mapping:

Course Outcome / Programme Outcomes	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12
CO2 - Apply the concepts of design to shafts, keys and couplings.	3	2	1	2	1	-	2	2	2